AMPHOTERICIN B

High-alert drug

Check formulation — doses differ

CHEMICAL NAME Polyene macrolide antifungal from *Streptomyces nodosus* (C47H73NO17); conventional deoxycholate or lipid formulations

CLASS Polyene antifungal (systemic)

TRADE NAMES Fungizone (conventional) · AmBisome (liposomal) · Abelcet (lipid complex)

THERAPEUTIC USES / INDICATIONS

- Life-threatening systemic fungal infections: invasive aspergillosis, candidaemia, mucormycosis.
- Cryptococcal meningitis (with flucytosine).
- Visceral leishmaniasis (kala-azar) (liposomal).
- Febrile neutropenia not responding to antibiotics (empirical).

ADMINISTRATION

- **IV infusion only** in 5% dextrose (precipitates in saline); flush line with dextrose.
- Conventional: infuse over 2–6 h; liposomal: 30–60 min (after 1st dose).
- Test dose (1 mg) may be given for conventional form.
- Premedicate (paracetamol, antihistamine, hydrocortisone) for infusion reactions; pre-hydration with saline (per protocol) reduces nephrotoxicity.

DOSAGES

- **Conventional:** 0.25–1 mg/kg/day — never exceed 1.5 mg/kg/day (fatal cardiac arrest).
- **Liposomal (AmBisome):** 3–5 mg/kg/day (mucormycosis up to 10 mg/kg).
- **Visceral leishmaniasis (liposomal):** 3 mg/kg on days 1–5, 14 and 21 (regimens vary).
- Confusion between formulations has caused deaths — confirm product name.

PHARMACOKINETICS

- Highly protein-bound; widely distributed; poor CSF penetration (still effective in meningitis).
- Slowly excreted in urine and bile over weeks.
- **t½:** 24 h initial, terminal ~15 days.

PHARMACODYNAMICS

- Binds **ergosterol** in fungal membrane → forms pores → leakage → fungicidal.
- Also binds cholesterol in human cells (→ toxicity, esp. kidney).
- Broad spectrum: *Candida*, *Aspergillus*, *Cryptococcus*, Mucorales, *Leishmania*.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Infusion reactions: fever, rigors, chills, nausea, headache.
- Thrombophlebitis.
- Anorexia, weight loss.

ADVERSE EFFECTS (SERIOUS)

- Nephrotoxicity (up to 80% with conventional form).
- Hypokalaemia, hypomagnesaemia (renal tubular wasting).
- Normocytic anaemia.
- Arrhythmias, cardiac arrest (overdose, rapid infusion).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity (unless life-threatening infection).
- **Caution:** renal impairment, concurrent nephrotoxic drugs.

KEY INTERACTIONS

- **Aminoglycosides, ciclosporin, tacrolimus, vancomycin:** ↑ nephrotoxicity.
- **Digoxin:** toxicity via hypokalaemia.
- **Corticosteroids, diuretics:** ↑ hypokalaemia.

NURSING MANAGEMENT

- Verify formulation and dose with second nurse.
- Use D5W only; protect from light during long infusions (conventional).
- Monitor vital signs every 30 min during infusion; treat rigors (pethidine may be prescribed).
- Daily creatinine, K^+, Mg^{2+}; weekly CBC, LFTs; fluid balance.
- Replace electrolytes as ordered.

PATIENT EDUCATION

- Fever and shivering during the drip are common — tell the nurse.
- Treatment may last weeks; blood tests will be done often.
- Report muscle weakness, palpitations or reduced urine.

KEY POINTS

- Polyene; broadest systemic antifungal.
- Dextrose only; infusion reactions.
- Nephrotoxic; causes hypokalaemia, hypomagnesaemia.
- Conventional max 1.5 mg/kg/day — check formulation.

SOURCES CHECKED **[239]** Amphotericin B Gilead liposomal 50 mg Powder for dispersion for infusion (previously known as AmBisome) (UK SmPC) **[240]** Amphotericin B, IV (US label) **[241]** Amphotericin B, IV (US label, Sun Pharmaceutical Industries)

Full references p. 432–457